

POSTS AND TELECOMMUNICATIONS INSTITUTE OF
TECHNOLOGY

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Chapter 2: MULTIPLE INTEGRALS

CALCULUS 2

Faculty of Fundamental Science 1

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1 2.1 Integral depends on a parameter

2 2.2 Double integrals

3 2.3. Triple integrals

2.1.1 Definite integral depends on a parameter

Definition 2.1

Let $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$, if for each fixed $y \in [c, d]$ the function $f(x, y)$ is integrable over $[a, b]$ on the x variable, we define the following function $F : [c, d] \rightarrow \mathbb{R}$ as

$$F(y) = \int_a^b f(x, y) dx$$

is called an integral depending on a parameter. The function $F(y)$ has the following properties:

Theorem 2.1

If the function $f(x, y)$ is continuous on $[a, b] \times [c, d]$ then $F(y)$ is continuous on $[c, d]$.

2.1.1 Definite integral depends on a parameter

Note 2.1

If $f(x, y)$ is continuous on $[a, b] \times [c, d]$, and $\alpha(y), \beta(y)$ are continuous on $[c, d]$ with $a \leq \alpha(y), \beta(y) \leq b$, $\forall y \in [c, d]$ then $F(y) = \int_{\alpha(y)}^{\beta(y)} f(x, y) dx$ is continuous on $[c, d]$.

Example 2.1

Let the function $f(x)$ be continuous on $[0, 1]$. Prove that

$$F(y) = \int_0^1 \frac{y^2 f(x)}{x^2 + y^2} dx$$

is continuous on $(0, +\infty)$.

2.1.1 Definite integral depends on a parameter

Theorem 2.2

If $f(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ are continuous on $[a, b] \times [c, d]$, then $F(y)$ is differentiable on $[c, d]$ and $F'(y) = \int_a^b \frac{\partial f}{\partial y}(x, y) dx$.

Theorem 2.3 (Leibniz's Theorem)

Let $f(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ be continuous functions on $[a, b] \times [c, d]$, and $\alpha(y), \beta(y)$ differentiable functions on $[c, d]$ with image on $[a, b]$, that is, $\alpha(y), \beta(y) : [c, d] \rightarrow [a, b]$, $\forall y \in [c, d]$. We define

$F(y) = \int_{\alpha(y)}^{\beta(y)} f(x, y) dx$, then $F(y)$ is differentiable on $[c, d]$ and

$$F'(y) = f(\beta(y), y) \cdot \beta'(y) - f(\alpha(y), y) \cdot \alpha'(y) + \int_{\alpha(y)}^{\beta(y)} \frac{\partial f}{\partial y}(x, y) dx.$$

2.1.1 Definite integral depends on a parameter

Example 2.2

Calculate the derivative of the following function

$$F(y) = \int_0^1 \arctan \frac{x}{y} dx, \quad y > 0.$$

Theorem 2.4

Let $f(x, y)$ be integrable over $[a, b] \times [c, d]$, then $F(y)$ is differentiable on $[c, d]$ and

$$\int_c^d F(y) dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx.$$

2.1.1 Definite integral depends on a parameter

Example 2.3

Calculating integrals

$$I = \int_0^1 \frac{x^b - x^a}{\ln x} dx, \quad b > a > 0.$$

2.1.2 Improper integral depends on parameter

Definition 2.2

1. Let $f : D := [a, +\infty) \times [c, d] \rightarrow \mathbb{R}$, if for each fixed $y \in [c, d]$ the function $f(x, y)$ is integrable over $[a, +\infty)$ on the x variable, we define

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

is called an improper integral of depending on a parameter of y .

2. The function $F(y)$ is called uniformly converge for each $y \in [c, d]$, if $\forall \varepsilon > 0, \exists n_0 = n_0(\varepsilon, y) > 0, \forall b \geq n_0 \Rightarrow \left| \int_b^{+\infty} f(x, y) dx \right| < \varepsilon$.
3. The function $F(y)$ is called uniformly converge on the interval $[c, d]$, if $\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}^*, \forall b \geq n_0 \Rightarrow \left| \int_b^{+\infty} f(x, y) dx \right| < \varepsilon, \forall y \in [c, d]$.

2.1.2 Improper integral depends on parameter

Theorem 2.5 (Weierstrass' theorem)

If the function $\int_a^{+\infty} h(x)dx$ converges and $|f(x, y)| \leq h(x)$, $\forall (x, y) \in D$ then the function $F(y)$ is uniformly converge on $[c, d]$.

Example 2.4

Prove that

$$\int_1^{+\infty} \frac{\cos(x + 2y)}{x^2 + y^2} dx$$

is continuous on $\mathbb{R}$.

2.1.2 Improper integral depends on parameter

Theorem 2.6

If the function $f(x, y)$ is continuous on $[a, +\infty) \times [c, d]$ and the function $F(y)$ is uniformly convergent on $[c, d]$ then $F(y)$ is continuous on $[c, d]$.

Example 2.5

Prove that

$$\int_1^{+\infty} \frac{x}{2+x^y} dx$$

is continuous on $(2, +\infty)$.

2.1.2 Improper integral depends on parameter

Theorem 2.7

If the function $f(x, y)$ is continuous on $[a, +\infty) \times [c, d]$ and the function $F(y)$ is uniformly convergent on $[c, d]$ then $F(y)$ is differentiable on $[c, d]$ and

$$\int_c^d F(y) dy = \int_c^d \left(\int_a^{+\infty} f(x, y) dx \right) dy = \int_a^{+\infty} \left(\int_c^d f(x, y) dy \right) dx.$$

Example 2.6

Given $b > a > 0$, calculate the following integral:

$$I = \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} dx.$$

2.1.2 Improper integral depends on parameter

Theorem 2.8

Let $f(x, y)$ be defined on the D satisfying the following assumptions:

- 1) the function $f(x, y)$ is continuous in the variable x on $[a, +\infty)$ for each $y \in [c, d]$,
- 2) a function $f'_y(x, y)$ is continuous in domain D ,
- 3) the function $F(y)$ converges for each $y \in [c, d]$,
- 4) a integral $\int_a^{+\infty} f'_x(x, y) dx$ converges uniformly on the $[c, d]$

Therefore $F'(y) = \int_a^{+\infty} f'_x(x, y) dx$.

Example 2.6

Find the derivative of the function

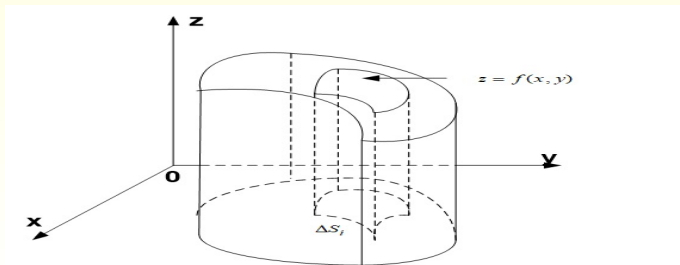
$$F(y) = \int_0^{+\infty} \frac{1 - \cos xy}{xe^{2x}} dx, y \in (0, +\infty), \text{ and find the function } F(y).$$

2.2.1 Definition of double integrals

Problem

Calculate the volume of the bounded domain V is given by:

- + (Oxy) is a plane.
- + The axis Oz and the standard curve L is the boundary of the finite closed domain $D \subset (Oxy)$.
- + The curved surface is the graph of a function of two variables $z = f(x, y), (x, y) \in D$.



2.2.1 Definition of double integrals

Definition

Let the function $z = f(x, y)$ define on a closed domain $D \subset R^2$.

- + Divide D into n small regions by a grid of curves, name and area the domains as $\Delta S_i (i = 1, \dots, n)$ and denoted d_i is the diameter of the second piece i .

- + Choose an arbitrary point $M_i (x_i, y_i) \in \Delta S_i$.

Then $I_n = \sum_{i=1}^n f(x_i, y_i) \Delta S_i$ is called the sum of the integrals of $f(x, y)$ on the domain D . D corresponds to the partition and how to choose the points $M_1, M_2, \dots, M_n$ as above when $n \rightarrow \infty$ so that $\max d_i \rightarrow 0$ but I_n does not depend on the partition ΔS_i and how to choose $M_i (x_i, y_i) \in \Delta S_i$ then number I is called the double integrals of $f(x, y)$ on the domain D and the symbol is

$$\iint_D f(x, y) dS \quad \text{So} \quad \iint_D f(x, y) dS = I = \lim_{\max d_i \rightarrow 0} \sum_{i=1}^n f(x_i, y_i) \Delta S_i$$

2.2.1 Definition of double integrals

Note

- 1) Since the double integral does not depend on the division of the domain D should be able to divide D by a grid of lines parallel to the coordinate axes Ox, Oy . Then $dS = dx \cdot dy$. Therefore, the double integral is denoted by

$$I = \iint_D f(x, y) dx dy$$

- 2) Like definite integrals, the symbol of a variable that is double integrated does not change the double integral, that is,

$$\iint_D f(x, y) dx dy = \iint_D f(u, v) du dv = I.$$

2.2.2 Integral conditions of double integrals

- If the function $f(x, y)$ is integrable over the domain D then $f(x, y)$ is bounded on the domain D (necessary condition of integrable function).
- If the function $f(x, y)$ is continuous on the D , more general: If the function $f(x, y)$ is only a discontinuity of type 1 on domain D , then it is integrable on the domain D .

2.2.3 Properties of double integrals

Let $f(x, y), g(x, y)$ be integrable on D . Then, we have

- 1) $\iint_D [f(x, y) \pm g(x, y)] dx dy = \iint_D f(x, y) dx dy \pm \iint_D g(x, y) dx dy.$
- 2) $\iint_D k \cdot f(x, y) dx dy = k \iint_D f(x, y) dx dy, \forall k.$
- 3) If $D = D_1 \cup D_2$ and $D_1 \cap D_2 = \emptyset$ then

$$\iint_D f(x, y) dx dy = \iint_{D_1} f(x, y) dx dy + \iint_{D_2} f(x, y) dx dy$$

2.2.3 Properties of double integrals

4) If $f(x, y) \leq g(x, y), \forall (x, y) \in D$ then

$$\iint_D f(x, y) dx dy \leq \iint_D g(x, y) dx dy$$

5) , If $f(x, y)$ is integral on D then $|f(x, y)|$ is also integrable on D and

$$\left| \iint_D f(x, y) dx dy \right| \leq \iint_D |f(x, y)| dx dy$$

6) If $f(x, y)$ is integral on D and satisfies $m \leq f(x, y) \leq M, \forall (x, y) \in D$ then

$$mS \leq \iint_D f(x, y) dx dy \leq MS.$$

where S is the area of the domain D .

2.2.4 Double integrals over rectangles

Theorem 2.2.1 (Fubini's Theorem)

Let $f(x, y)$ be continuous on $D = [a, b] \times [c, d]$ (the domain D is a rectangular domain). We have

$$\iint_D f(x, y) dx dy = \int_a^b dx \left(\int_c^d f(x, y) dy \right) = \int_c^d dy \left(\int_a^b f(x, y) dx \right).$$

Example 2.2.1

Calculating integrals $I = \iint_D (2x + y) dx dy$, where $D = [1, 2] \times [0, 2]$.

Example 2.2.2

Calculating integrals $I = \iint_D xy^2 dx dy$, where $D = [0, 2] \times [0, 3]$.

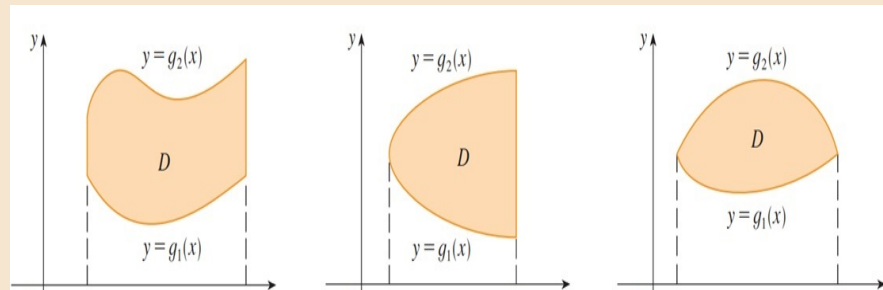
2.2.4 Double integrals over general regions

Type I regions

A plane region D is said to be of type I if it lies between the graphs of two continuous functions of x , that is

$D = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$. If $f(x, y)$ is continuous on a type I region D then

$$I = \iint_D f(x, y) dx dy = \int_a^b dx \left(\int_{g_1(x)}^{g_2(x)} f(x, y) dy \right).$$

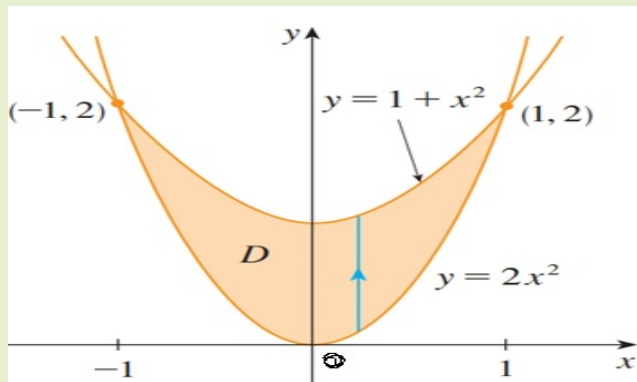


2.2.4 Double integrals over general regions

Example 2.2.3

Evaluate $I = \iint_D (x^2 + 2y) dx dy$, where D is a region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Solution



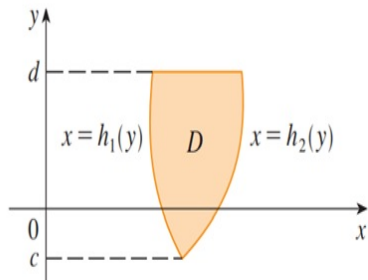
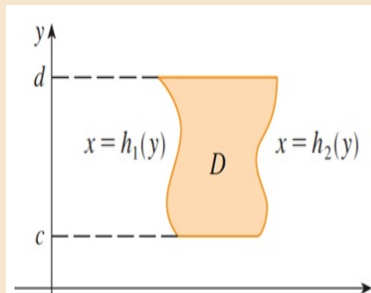
2.2.4 Double integrals over general regions

Type II regions

A plane region D is said to be of type II if it lies between the graphs of two continuous functions of y , that is

$D = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$. If $f(x, y)$ is continuous on a type II region D then

$$I = \iint_D f(x, y) dx dy = \int_c^d dy \left(\int_{h_1(y)}^{h_2(y)} f(x, y) dx \right).$$

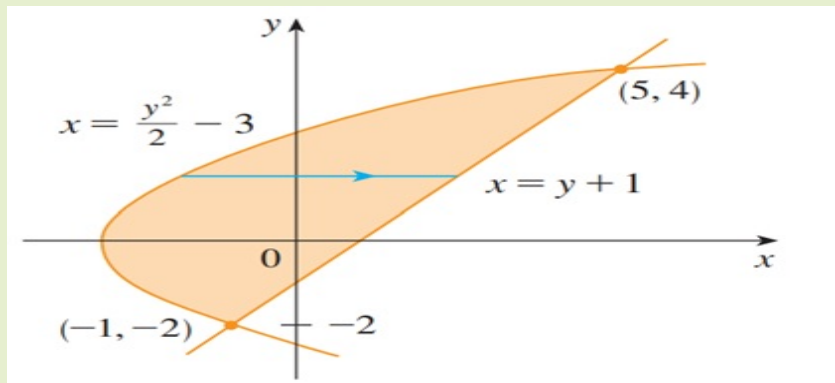


2.2.4 Double integrals over general regions

Example 2.2.4

Evaluate $I = \iint_D xy dx dy$, where D is a region bounded by the parabolas $y = x - 1$ and $y^2 = 2x + 6$.

Solution



2.2.4 Double integrals over general regions

Example 2.2.5

Change the order of integration in double integrals

$$a) I = \int_0^2 dx \int_x^{2x} f(x, y) dy.$$

$$b) J = \int_{-2}^6 dy \int_{-\frac{y^2}{2}-1}^{2-y} f(x, y) dx.$$

$$c) K = \int_0^1 dx \int_x^{\sqrt{2-x^2}} f(x, y) dy.$$

2.2.5 Change of variables in double integrals

Let the function $f(x, y)$ be continuous on the domain $D \subset (Oxy)$ and assume the transformation $(x, y) \rightarrow (u, v) : \begin{cases} x = x(u, v) \\ y = y(u, v) \end{cases}$ satisfying the condition

- The above transformation is a bijective from Δ to the domain D or $(x, y) \in D \Leftrightarrow (u, v) \in \Delta$.
- The $x(u, v), y(u, v)$ are the continuous partial derivatives on the domain $\Delta \subset (O'uv)$.
- The Jacobi determinant is $\frac{D(x, y)}{D(u, v)} \neq 0$ on the domain Δ (or just zero at some isolated point) then

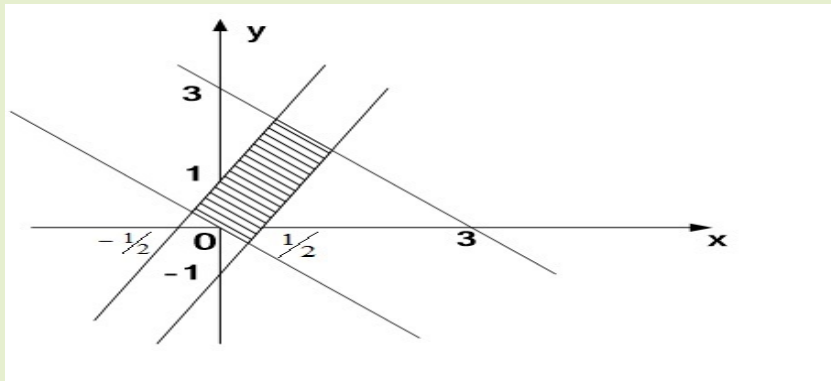
$$I = \iint_D f(x, y) dx dy = \iint_{\Delta} f[x(u, v), y(u, v)] \cdot \left| \frac{D(x, y)}{D(u, v)} \right| du dv.$$

2.2.5 Change of variables in double integrals

Example 2.2.6

Calculating integrals $I = \iint_D (x + y) dx dy$, where D is $y = -x, y = -x + 3, y = 2x - 1, y = 2x + 1$.

Solution



2.2.5 Change of variables in double integrals

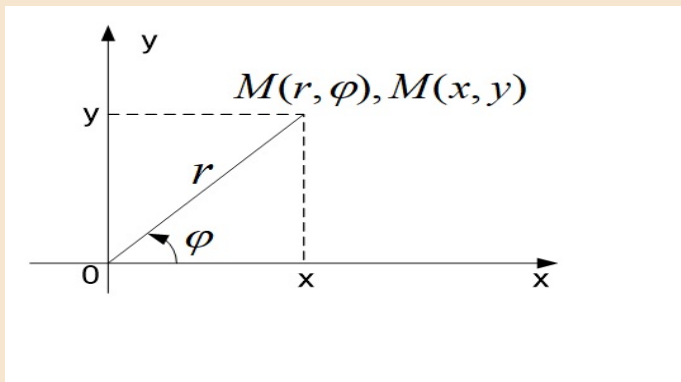
Example 2.2.7

Calculating integrals $I = \iint_D x^3 dx dy$, where D is
 $y = \frac{1}{x}, y = \frac{2}{x}, y = x^2, y = \frac{x^2}{2}$.

2.2.6 Double integrals in polar coordinates

a. polar coordinate system

A polar coordinates are set of real numbers (r, φ) so that
 $r = |\overrightarrow{OM}|, \varphi = (Ox, \overrightarrow{OM})$



2.2.6 Double integrals in polar coordinates

b. calculate the double integrals

Set

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} \Leftrightarrow \begin{cases} r = \sqrt{x^2 + y^2} \\ \tan \varphi = \frac{y}{x} \end{cases} \Rightarrow D \rightarrow \Delta = \left\{ (r, \varphi) \mid \begin{cases} 0 \leq \varphi < 2\pi \\ 0 \leq r < +\infty \end{cases} \right.$$

$$J = \frac{D(x, y)}{D(r, \varphi)} = \begin{vmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{vmatrix} = r$$

Then the double integrals in polar coordinates has the form

$$I = \iint_D f(x, y) dx dy = \iint_{\Delta} f(r \cos \varphi, r \sin \varphi) r dr d\varphi.$$

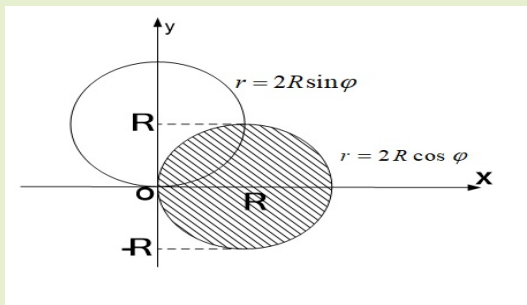
2.2.6 Double integrals in polar coordinates

Example 2.2.8

Calculate $I = \iint_D \sqrt{x^2 + y^2} dx dy$, where, the domain D is defined by

$$D = \{(x, y) : x^2 + y^2 \leq 2Ry, x^2 + y^2 \geq 2Rx\}.$$

Solution



2.2.7 Applications of double integrals

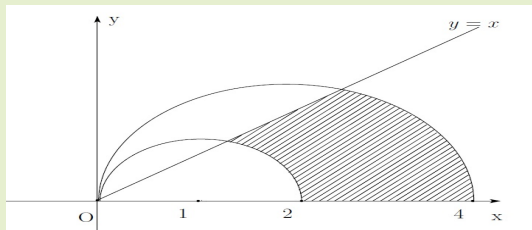
1. Finding area of a plane regions

If $f(x, y) = 1$, $\forall (x, y) \in D$ then the measure of the area of the domain D is calculated according to the formula $S_D = \iint_D dx dy$.

Example 2.2.9

Calculate the area of the domain D given by

$$D = \{(x, y) : (x - 1)^2 + y^2 = 1, (x - 2)^2 + y^2 = 4, y = x, y = 0\}.$$



2.2.7 Applications of double integrals

2. Computing volumes

If $f(x, y) \geq 0$, $\forall (x, y) \in D$ then the volume of the curved cylinder bounded by the function graph is calculated by the formula

$$V = \iint_D f(x, y) dx dy.$$

Example 2.2.10

Calculate the volume of the figure V given by the following faces

$$z = x^2 + y^2, y = x^2, y = 1, z = 0.$$

2.2.7 Applications of double integrals

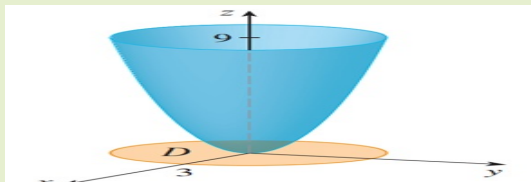
3. Surface area

For surface $(S) : z = f(x, y)$, $(x, y) \in D$ has partial derivatives f'_x, f'_y exist and are continuous on domain D . Then the surface area of S is defined as

$$A(S) = \iint_D \sqrt{1 + f'^2_x + f'^2_y} dx dy.$$

Example 2.2.11

Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 9$.



2.2.7 Applications of double integrals

4. Density mass

Suppose the lamina occupies a region D of the xy -plane and its density (in units of mass per unit area) at a point (x, y) in D is given by $\rho(x, y)$, where ρ is a continuous function on D . The density mass of the lamina is

$$m = \iint_D \rho(x, y) dx dy.$$

If the plate is homogenous, that is $\rho(x, y) = \text{const}, \forall (x, y) \in D$, choose $\rho(x, y) = 1, \forall (x, y) \in D$ then the mass of the plate D is calculated by the formula $m = \iint_D dx dy = S_D$

5. Center of mass

$$x_G = \frac{1}{m} \iint_D x \rho(x, y) dx dy, \quad y_G = \frac{1}{m} \iint_D y \rho(x, y) dx dy,$$

where $m = \iint_D \rho(x, y) dx dy$.

2.2.7 Applications of double integrals

4. Density mass

Suppose the lamina occupies a region D of the xy -plane and its density (in units of mass per unit area) at a point (x, y) in D is given by $\rho(x, y)$, where ρ is a continuous function on D . The density mass of the lamina is

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If the plate is homogenous, that is $\rho(x, y) = \text{const}, \forall (x, y) \in D$, choose $\rho(x, y) = 1, \forall (x, y) \in D$ then the mass of the plate D is calculated by the formula $m = \iint_D dx dy = S_D$

5. Center of mass

$$x_G = \frac{1}{m} \iint_D x \rho(x, y) dx dy, \quad y_G = \frac{1}{m} \iint_D y \rho(x, y) dx dy,$$

where $m = \iint_D \rho(x, y) dx dy$.

2.2.7 Applications of double integrals

6. Moment of inertia

According to the definition of the moment of inertia of the particle about the Ox, Oy -axis and the origin O , we have

$$I_{Ox} = my^2; \quad I_{Oy} = mx^2; \quad I_O = m(x^2 + y^2)$$

Moment of inertia of the plate about the axes Ox, Oy and the origin O are

$$I_{Ox} = \iint_D y^2 \rho(x, y) dx dy; \quad I_{Oy} = \iint_D x^2 \rho(x, y) dx dy;$$

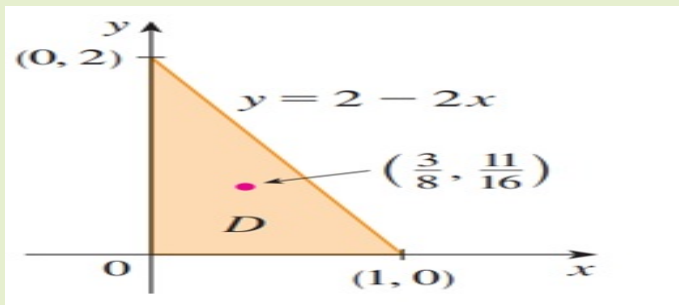
$$I_O = \iint_D (x^2 + y^2) \rho(x, y) dx dy.$$

2.2.7 Applications of double integrals

Example 2.2.12

Find the mass and center of mass of a triangular lamina with vertices $(0, 0)$, $(1, 0)$, and $(0, 2)$ if the density function is $\rho(x, y) = 1 + 3x + y$.

Solution



2.3.1 Definition of triple integrals

Problem

Calculate the mass of the non-homogeneous body V , given that the density is $\rho = \rho(x, y, z)$, $(x, y, z) \in V$. Similar to the double integral, we divide V arbitrarily into n parts that do not step on each other. Name and volume of the parts $\Delta V_i (i = \overline{1, n})$. Choose an arbitrary point $P_i (x_i, y_i, z_i) \in \Delta V_i$ and the $d_i, (i = \overline{1, n})$ are diameters of $\Delta V_i (i = \overline{1, n})$. We have

$$m \approx \sum_{i=1}^n \rho(P_i) \Delta V_i = \sum_{i=1}^n \rho(x_i, y_i, z_i) \Delta V_i$$

The mass of the object is

$$m = \lim_{\max d_i \rightarrow 0} \sum_{i=1}^n \rho(x_i, y_i, z_i) \Delta V_i$$

2.3.1 Definition of triple integrals

Definition

Let the function $f(x, y, z)$ define on the domain $V \subset \mathbb{R}^3$.

- Divide V into n pieces, name and volume of the piece are $\Delta V_i (i = \overline{1, n})$, the piece diameter symbol ΔV_i is $d_i, i = \overline{1, n}$.
- Choose an arbitrary point $P_i (x_i, y_i, z_i) \in \Delta V_i, (i = \overline{1, n})$.
- The totals $I_n = \sum_{i=1}^n f(x_i, y_i, z_i) \Delta V_i$ is called the sum of integrals the triple of the function $f(x, y, z)$ taken over the domain V corresponds to a fraction plan and points $P_i \in \Delta V_i, (i = \overline{1, n})$..

When $n \rightarrow \infty$ such that $\max d_i \rightarrow 0$, we get I_n converges to $I \in \mathbb{R}$ regardless of the partition ΔV_i and how point $P_i (x_i, y_i, z_i) \in \Delta V_i$ is chosen, the number I is called a triple integral of $f(x, y, z)$ over the region V and denoted by

$$\iiint_V f(x, y, z) dV = I = \lim_{\max_i d_i \rightarrow 0} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta V_i$$

2.3.1 Definition of triple integrals

Note

- Like the double integrals, the volume factor dV is replaced by $dx dy dz$ and then the triple integral is usually denoted by

$$I = \iiint_V f(x, y, z) dx dy dz$$

- Similar to the double integrals, triple integrals do not depend on the notation of the variable being integrated

$$\iiint_V f(x, y, z) dx dy dz = \iiint_V f(u, v, w) du dv dw$$

- If the function $f(x, y, z)$ is continuous on the closed, bounded domain $V \in \mathbb{R}^3$, then it's integrable on V .
- The integral conditions and properties of triple integrals are similar to the double integrals.

2.3.2 Triple integrals on the rectangular box

Theorem 2.3.1 (Fubini's Theorem)

If $f(x, y, z)$ is continuous on the rectangular box
 $V = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_V f(x, y, z) dx dy dz = \int_a^b \int_c^d \int_r^s f(x, y, z) dx dy dz.$$

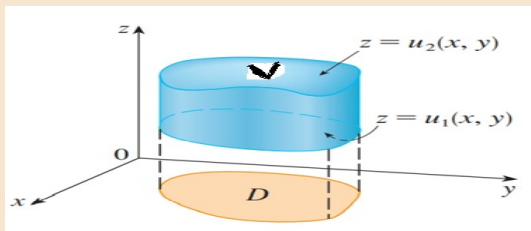
Example 2.3.1

Calculating integrals $I = \iiint_V xyz^2 dx dy dz$, where

$$V = [0, 1] \times [-1, 2] \times [0, 3].$$

2.3.2 The triple integral over a general bounded region

The region of type I



A solid region V is said to be of **type 1** if it lies between the graphs of two continuous functions of x and y , that is,

$V = \{(x, y, z) | (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$, then

$$I = \iiint_V f(x, y, z) dx dy dz = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right] dx dy.$$

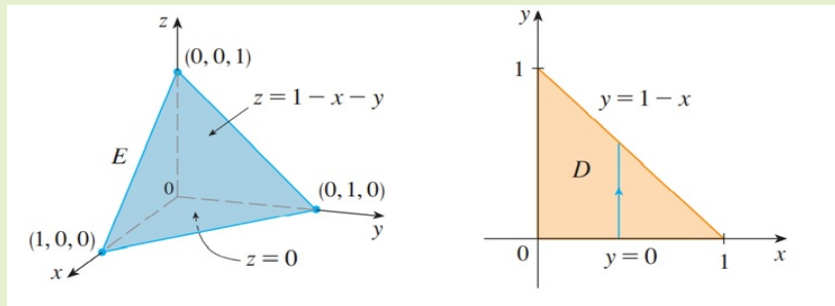
2.3.2 The triple integral over a general bounded region

where D is the projection of V onto xy -plane, $z = u_1(x, y)$ is the lower surface and $z = u_2(x, y)$ is the upper surface.

Example 2.3.2

Evaluate $I = \iiint_V z dx dy dz$, where V is the solid tetrahedron bounded by the four planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

Solution

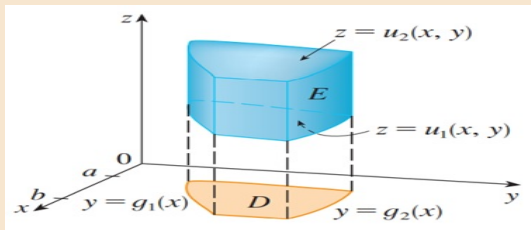


2.3.2 The triple integral over a general bounded region

The region of type II

If the projection D of V onto the xy -plane is of type II plane region, then

$V = \{(x, y, z) | a \leq x \leq b, g_1(x) \leq y \leq g_2(x), u_1(x, y) \leq z \leq u_2(x, y)\}$,
and we have



$$I = \iiint_V f(x, y, z) dx dy dz = \int_a^b dx \int_{g_1(x)}^{g_2(x)} dy \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz.$$

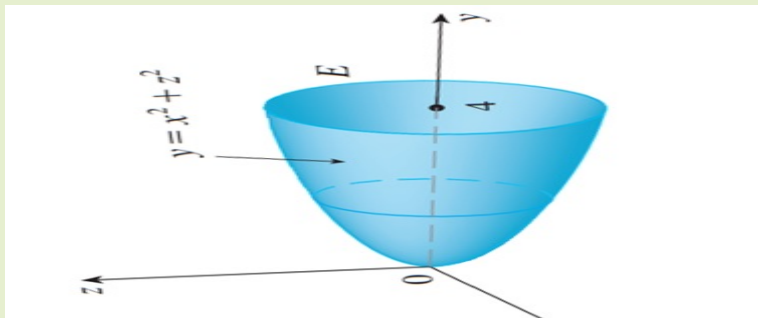
2.3.2 The triple integral over a general bounded region

where D is the projection of V onto xy -plane, $z = u_1(x, y)$ is the lower surface and $z = u_2(x, y)$ is the upper surface.

Example 2.3.3

Evaluate $\iiint_V \sqrt{x^2 + z^2} dx dy dz$, where V is the solid tetrahedron bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

Solution



2.3.3 Change of variables in triple integrals

For the function $f(x, y, z)$ to be continuous on the domain $V \subset Oxyz$ and and assume the transformation

$$(x, y, z) \rightarrow (u, v, w) : \begin{cases} x = x(u, v, w) \\ y = y(u, v, w), & (u, v, w) \in \Omega \text{ satisfy the} \\ z = z(u, v, w) \end{cases}$$

conditions

- The above transformation is a bijective from Ω to the domain V or $(x, y, z) \in V \Leftrightarrow (u, v, w) \in \Omega$.
- The $x(u, v, w), y(u, v, w), z(u, v, w)$ are the continuous partial derivatives on the domain $\Omega \subset (O'uvw)$.
- The Jacobi determinant is $J = \frac{D(x, y, z)}{D(u, v, w)} \neq 0$ on the domain Ω , then

$$\iiint_V f(x, y, z) dx dy dz = \iiint_{\Omega} f[x(u, v, w), y(u, v, w), z(u, v, w)] |J| du dv dw$$

2.3.3 Change of variables in triple integrals

Example 2.3.4

Evaluate $\iiint_V (x+y)(x-z) dx dy dz$, where V is the bounded domain by the planes

$$x+y=0, x+y=1; y+z=1, y+z=2; x+y-z=2, x+y-z=3..$$

Solution

Set

$$u = x + y, v = y + z, w = x + y - z$$

$$0 \leq u \leq 1, 1 \leq v \leq 2, 2 \leq w \leq 3$$

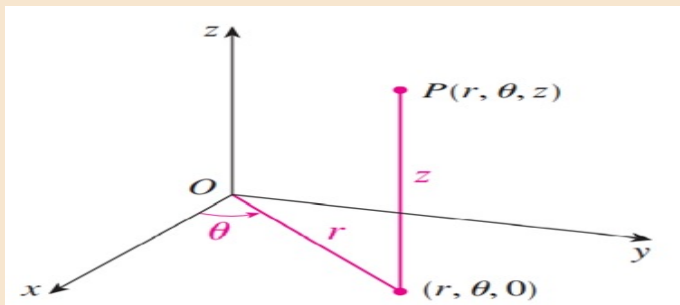
$$\frac{D(u, v, w)}{D(x, y, z)} = -1 \Rightarrow \frac{D(x, y, z)}{D(u, v, w)} = -1, (x+y)(x-z) = u(u-v)$$

$$I = \iiint_{\Omega} u(u-v) | -1 | du dv dw = \int_0^1 u du \int_1^2 (u-v) dv \int_2^3 dw = -\frac{5}{12}.$$

2.3.4 Triple integrals in cylindrical coordinates

Cylindrical coordinates

In the cylindrical coordinate system, a point P in three-dimensional space is represented by the ordered triple (r, θ, z) , where r and θ are polar coordinates of the projection of P onto the xy -plane and z is the directed distance from the xy -plane. The connections between cylindrical coordinates and rectangular coordinates are



2.3.4 Triple integrals in cylindrical coordinates

Evaluating triple integrals with cylindrical coordinates

$$\text{Set } \begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \\ z = z \end{cases} \text{ then } V \rightarrow \Omega : \begin{cases} r \geq 0 \\ 0 \leq \varphi < 2\pi \\ -\infty < z < +\infty \end{cases}.$$

The Jacobi determinant of the functions x, y, z in terms of r, φ, z are

$$J = \frac{D(x, y, z)}{D(r, \varphi, z)} = \begin{vmatrix} \cos \varphi & -r \sin \varphi & 0 \\ \sin \varphi & r \cos \varphi & 0 \\ 0 & 0 & 1 \end{vmatrix} = r.$$

The formula for triple integration in cylindrical coordinates is

$$I = \iiint_V f(x, y, z) dx dy dz = \iiint_{\Omega} f(r \cos \varphi, r \sin \varphi, z) r dr d\varphi dz.$$

2.3.4 Triple integrals in cylindrical coordinates

Example 2.3.5

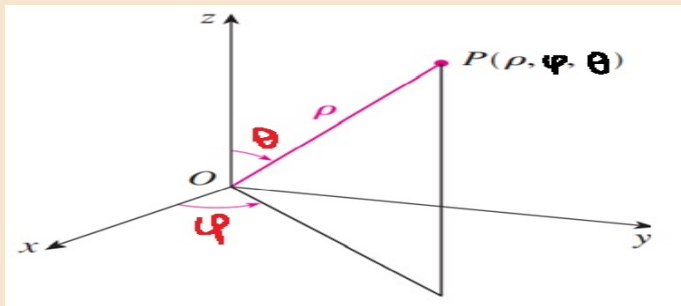
Evaluate $I = \iiint_V (x^2 + y^2 + 3z^2) dx dy dz$, where

$$V = \{(x, y, z) | \sqrt{x^2 + y^2} \leq z \leq 2\}.$$

2.3.5 Triple integrals in spherical coordinates

Spherical coordinates

The spherical coordinates of a point P in space are (ρ, φ, θ) , where ρ is the distance from P to the origin, φ is the same angle as in cylindrical coordinates, and θ is the angle between the positive z -axis and the line segment OP . Note that $\rho \geq 0$, $0 \leq \theta \leq \pi$.



2.3.5 Triple integrals in spherical coordinates

Evaluating triple integrals with spherical coordinates

$$\text{Set } \begin{cases} x = r \cos \varphi \cos \theta \\ y = r \cos \varphi \sin \theta \\ z = r \sin \theta \end{cases} \text{ then } V \rightarrow \Omega : \begin{cases} r \geq 0 \\ 0 \leq \varphi < 2\pi \\ 0 \leq \theta \leq \pi \end{cases}.$$

The Jacobi determinant of the functions x, y, z in terms of r, φ, θ are

$$J = \frac{D(x, y, z)}{D(r, \varphi, \theta)} = \begin{vmatrix} \sin \theta \cos \varphi & -r \sin \theta \sin \varphi & r \cos \theta \cos \varphi \\ \sin \theta \sin \varphi & r \sin \theta \cos \varphi & r \cos \theta \sin \varphi \\ \cos \theta & 0 & -r \sin \theta \end{vmatrix} = -r^2 \sin \theta.$$

The formula for triple integration in spherical coordinates is

$$I = \iiint_{\Omega} f(r \cos \varphi \cos \theta, r \cos \varphi \sin \theta, r \sin \theta) r^2 \sin \theta dr d\varphi d\theta.$$

2.3.5 Triple integrals in spherical coordinates

Evaluating triple integrals with spherical coordinates

$$\text{Set } \begin{cases} x = r \cos \varphi \cos \theta \\ y = r \cos \varphi \sin \theta \\ z = r \sin \theta \end{cases} \text{ then } V \rightarrow \Omega : \begin{cases} r \geq 0 \\ 0 \leq \varphi < 2\pi \\ 0 \leq \theta \leq \pi \end{cases}.$$

The Jacobi determinant of the functions x, y, z in terms of r, φ, θ are

$$J = \frac{D(x, y, z)}{D(r, \varphi, \theta)} = \begin{vmatrix} \sin \theta \cos \varphi & -r \sin \theta \sin \varphi & r \cos \theta \cos \varphi \\ \sin \theta \sin \varphi & r \sin \theta \cos \varphi & r \cos \theta \sin \varphi \\ \cos \theta & 0 & -r \sin \theta \end{vmatrix} = -r^2 \sin \theta.$$

The formula for triple integration in spherical coordinates is

$$I = \iiint_{\Omega} f(r \cos \varphi \cos \theta, r \cos \varphi \sin \theta, r \sin \theta) r^2 \sin \theta dr d\varphi d\theta.$$

2.3.5 Triple integrals in spherical coordinates

Example 2.3.6

Evaluate $I = \iiint_V \sqrt{x^2 + y^2 + z^2} dx dy dz$, where

- a) V is the unit ball.
- b) $V = \{(x, y, z) | x^2 + y^2 + z^2 \leq 4\}$.
- c) $V = \{(x, y, z) | 0 \leq z \leq \sqrt{4 - x^2 - y^2}\}$.

2.3.6 Applications of triple integrals

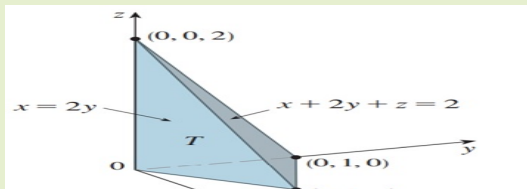
1. Volume

If $f(x, y, z) = 1$ for all point in V . Then the triple integral does represent the volume of V

$$V = \iiint_V dV = \iiint_V dx dy dz.$$

Example 2.3.7

Use triple integral to find the volume of the tetrahedron V bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.



2.3.6 Applications of triple integrals

2. Mass of a solid object

If the density function of a solid object that occupies the region V is $\rho(x, y, z)$ (in units of mass per unit volume) at any given point (x, y, z) , then its mass is

$$m = \iiint_V \rho(x, y, z) dV = \iiint_V \rho(x, y, z) dx dy dz.$$

2.3.6 Applications of triple integrals

3. Moments

Its moments about the three coordinate planes are

$$I_{Ox} = \iiint_V (y^2 + z^2) \rho(x, y, z) dV, \quad I_{Oy} = \iiint_V (x^2 + z^2) \rho(x, y, z) dV$$

$$I_{Oz} = \iiint_V (x^2 + y^2) \rho(x, y, z) dV, \quad I_O = \iiint_V (x^2 + y^2 + z^2) \rho(x, y, z) dV.$$

2.3.6 Applications of triple integrals

4. Center of mass

The center of mass is located at the point G , where

$$x_G = \frac{1}{m} \iiint_V x \rho(x, y, z) dV, \quad y_G = \frac{1}{m} \iiint_V y \rho(x, y, z) dV$$

$$z_G = \frac{1}{m} \iiint_V z \rho(x, y, z) dV, \quad m = \iiint_V \rho(x, y, z) dV.$$

2.3.6 Applications of triple integrals

Example 2.3.8

Find the center of mass (if the density is constant, the center of mass is called the centroid) of a solid of constant density that is bounded by the parabolic cylinder $x = y^2$ and the planes $x = z$, $z = 0$, and $x = 1$.